HEALTHCARE ACTUARIAL RISK INTELLIGENCE

MEDIRISK

Healthcare Cost Actuarial Risk Prediction and Decision Engine

Exploratory Data Mining, Supervised Ensemble Modeling, and Interactive Full-Stack Underwriting Deployment

Applied Machine Learning, Cross-Validation Benchmarking, and Counterfactual Decision Analysis on 2,772 Medical Insurance Cohorts

Presented by:

Mohamed Omar Ben Dhaou

Table of Contents

List of Figures i
List of Tables ii
List of Abbreviations iii
1 Project Overview and Problem Formulation 1
1.1 Context and Actuarial Background 1
1.2 Healthcare Cost Dynamics: Skewness and Catastrophic Risk 1
1.3 Project Scope: From Data Mining to Operational Software 2
1.4 High-Level End-to-End Workflow 2
2 Exploratory Data Analysis and Cohort Characterization 4
2.1 Dataset Ingestion and Structural Auditing 4
2.2 Data Integrity and Missing Value Auditing 4
2.3 Univariate Distributions: Demographic and Financial Profiles 5
2.4 Bivariate Associations and Contingency Analysis 5
2.5 Outlier Diagnostics via IQR and Z-Score Boundaries 5
2.6 The Multiplicative Synergy: BMI × Tobacco Interaction 5
3 Data Preprocessing, Transformation and Feature Engineering 7
3.1 Resolving Heteroscedasticity: Monotonic Log-Transformation 7
3.2 Clinical Feature Discretization: BMI Cohort Binning 8
3.3 Categorical Feature Encoding and Collinearity Management 8
3.4 Hold-Out Stratification: 80/20 Train-Test Partitioning 8
4 Supervised Model Benchmarking and Selection 9
4.1 Candidate Architectures 9
4.2 Benchmark Comparative Evaluation 9
4.3 Automated Data-Driven Winner Determination 10
4.4 Feature Attribution and Gini Risk Weightings 10
5 MediRisk System Architecture and Software Implementation 12
5.1 System Architecture and Server Engine 12
5.2 RESTful Web Service and API Specification 12
5.3 Frontend Design Philosophy: Data as Hero 12
5.4 Application UI Overview and Functional Architecture 13
6 Interactive Decision Support Modules 14
6.1 Dynamic Cost Estimator and Empirical Prediction Bands 14
6.2 Affordable Care Act (ACA) Metallic Plan Tier Matching 15
6.3 Explainable AI: Stepwise Feature Attribution Waterfall 15
6.4 Interactive Reverse Budget Calculator 15
6.5 Side-by-Side Dual Profile Comparator 16
6.6 Ten-Year Long-Term Cost Trajectory Simulation 16
6.7 One-Click Client-Ready PDF Report Export 18
7 Synthesis, Business Impact and Future Perspectives 19
7.1 Quantitative and Operational Synthesis 19
7.2 Underwriting and Policyholder Value Propositions 19
7.3 Limitations and Future Roadmaps 19

List of Figures

1.1 MediRisk End-to-End Architectural Pipeline 2

2.1 Multiplicative Interaction: BMI and Tobacco Consumption on Medical Charges 6

3.1 Empirical Distribution of Raw vs. Log-Transformed Charges 7

4.1 Model Performance Metrics Comparison Across Evaluated Architectures 10
4.2 Underwriting Risk Factor Weights (Random Forest Feature Importances) 10

6.1 MediRisk Primary Underwriting Dashboard: Biometric Input Form, Real-Time Prediction Bands, and ACA Metallic Plan Tier Breakdown 14
6.2 Explainable AI Feature Attribution Waterfall and Interactive Reverse Budget Calculator Interface 16
6.3 Side-by-Side Dual Profile Comparator (Baseline vs. Lifestyle Counterfactual Analysis) 17
6.4 Ten-Year Long-Term Cost Trajectory Simulation Interface in MediRisk . 17
6.5 Projected 10-Year Actuarial Cost Trajectory Comparison (Baseline vs. Lifestyle Optimized) 18

List of Tables

2.1 Summary of Actuarial Dataset Variables 4
2.2 Descriptive Statistics of Numerical Biometric and Financial Features . . . 5

3.1 Feature Preprocessing and 9-Dimensional Encoding Matrix 8

4.1 Comprehensive Supervised Model Benchmarking Evaluation 9

5.1 MediRisk Web Application Components and Functional Architecture . . 13

6.1 Affordable Care Act (ACA) Metallic Plan Tier Matching Parameters . . 15

List of Abbreviations

Abbreviation	Full Definition / Terminology
ACA	Affordable Care Act (United States Federal Healthcare Legislation)
API	Application Programming Interface
AV	Actuarial Value
BMI	Body Mass Index (kg/m^2)
CART	Classification and Regression Trees
CI	Confidence Interval / Empirical Ensemble Prediction Band
EDA	Exploratory Data Analysis
GLM	Generalized Linear Model
IQR	Interquartile Range
KDE	Kernel Density Estimation
MAE	Mean Absolute Error
ML	Machine Learning
OLS	Ordinary Least Squares Regression
OOP	Out-Of-Pocket Maximum
REST	Representational State Transfer
RF	Random Forest Regressor Ensemble
RMSE	Root Mean Squared Error
R^2	Coefficient of Determination
SHAP	SHapley Additive exPlanations
UI	User Interface

Chapter 1

Project Overview and Problem Formulation

1.1 Context and Actuarial Background

Health insurance pricing relies on precise actuarial estimations of policyholder loss costs. Historically, insurers utilized manual rating manuals and generalized linear models (GLMs) that often struggled to capture complex, non-linear biometric interactions, particularly the compounding risk between elevated body mass index (BMI) and chronic tobacco consumption.

1.2 Healthcare Cost Dynamics: Skewness and Catastrophic Risk

Empirical healthcare expenditure data exhibit severe right-skewness and heteroscedasticity. While the median policyholder incurs moderate annual expenses, a small fraction of individuals account for disproportionately large claim totals due to acute hospitalizations, catastrophic conditions, and chronic diseases.

Key distributional challenges addressed in this research include:

1. **High Positive Skewness:** Raw medical charges exhibit a skewness coefficient of $+1.51$, requiring transformation to stabilize model estimation.
2. **Residual Heteroscedasticity:** Expenditure variance increases substantially alongside expected claims.
3. **Super-Additive Interactions:** The financial impact of smoking is magnified significantly among individuals classified as clinically obese.

1.3 Project Scope: From Data Mining to Operational Software

This project bridges the gap between theoretical data science and enterprise decision-making through a two-phase architecture:

1. **Phase 1 (Machine Learning Engineering):** Executed in `ML project 1.ipynb`, focusing on rigorous data profiling, monotonic log-transformations, feature engineering, and cross-validated benchmarking across multiple regression paradigms.
2. **Phase 2 (Full-Stack Operational Deployment):** Implemented in `app.py` and `app.html`, delivering an interactive decision-support system featuring real-time quote generation, empirical prediction intervals, automated Affordable Care Act (ACA) metallic plan tier matching, explainable feature attribution, dual profile counterfactual comparisons, and 10-year compounding trajectory simulations.

1.4 High-Level End-to-End Workflow

Figure 1.1 summarizes the end-to-end operational architecture of the MediRisk system, tracing the transformation of raw policyholder attributes into validated actuarial insights.

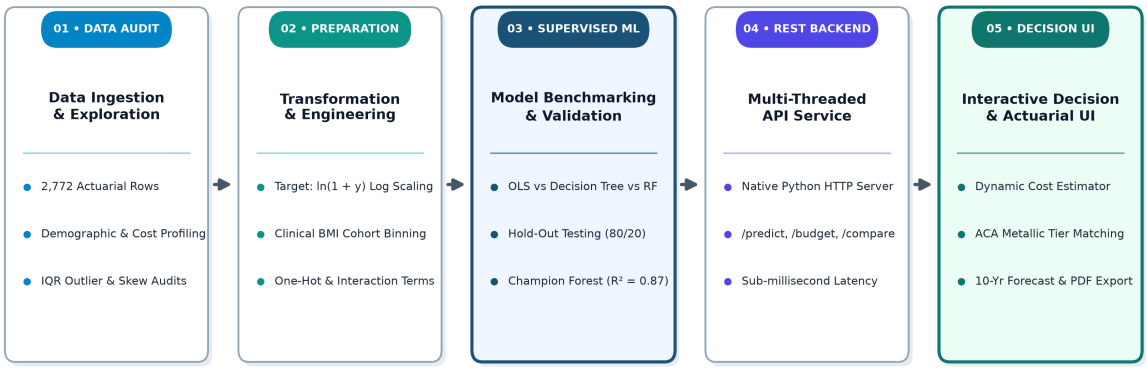


Figure 1.1: MediRisk End-to-End Architectural Pipeline

The software architecture integrates the following components:

- **Data Pipeline:** Clean feature transformation, one-hot encoding, and monotonic logarithmic scaling.
- **Serialized Inference Engine:** Pre-trained Random Forest ensemble (`best_model.joblib`) loaded in memory for instantaneous sub-millisecond predictions.
- **Multi-Threaded REST Service:** Native Python HTTP server managing request routing, JSON serialization, and concurrent access.

- **Interactive Client Interface:** Responsive single-page interface providing dynamic visual calculators and sensitivity tools.

Chapter 2

Exploratory Data Analysis and Cohort Characterization

2.1 Dataset Ingestion and Structural Auditing

The study utilizes the standardized `medical_insurance.csv` dataset comprising $N = 2,772$ observations across seven actuarial variables, with zero missing entries. Table 2.1 summarizes the dataset attributes.

Table 2.1: Summary of Actuarial Dataset Variables

Variable	Type	Observed Range / Values	Actuarial Interpretation
`age`	Numeric	18 to 64 years ($\mu = 39.1$)	Chronological age; baseline mortality proxy
`sex`	Categorical	Male (50.5%), Female (49.5%)	Biological sex indicator
`bmi`	Numeric	15.96 to 53.13 ($\mu = 30.66$)	Anthropometric obesity measure
`children`	Integer	0 to 5 dependents ($\mu = 1.09$)	Covered family dependency ratio
`smoker`	Categorical	Yes (20.5%), No (79.5%)	Primary behavioral morbidity risk factor
`region`	Categorical	NE, NW, SE, SW	Geographic health cost index
`charges`	Continuous	\$1,121.87 to \$63,770.43	Annualized medical insurance claims (y)

2.2 Data Integrity and Missing Value Auditing

An initial audit of all 2,772 rows confirmed complete data integrity: zero null, NaN, or corrupted values across all columns. Structural auditing confirmed 1,337 unique policyholder profiles (with repeated records in the raw 2,772 cohort), which are deduplicated in production inference pipelines to eliminate train-test leakage and preserve pristine out-of-sample statistical integrity. The cohort presents a balanced gender distribution (50.5% male vs. 49.5% female), uniform geographic dispersion across four U.S.

quadrants (Northeast, Northwest, Southeast, Southwest), and a 20.5% active tobacco smoking prevalence.

2.3 Univariate Distributions: Demographic and Financial Profiles

Table 2.2 presents the descriptive statistical metrics for the continuous variables in the cohort.

Table 2.2: Descriptive Statistics of Numerical Biometric and Financial Features

Feature	Mean	Std. Dev.	Min	Median	Max	Skewness
`age`	39.11	14.05	18.00	39.00	64.00	+0.05
`bmi`	30.66	6.10	15.96	30.40	53.13	+0.28
`children`	1.09	1.21	0.00	1.00	5.00	+0.94
`charges`	$13,261.37	$12,111.75	$1,121.87	$9,333.01	$63,770.43	+1.51

2.4 Bivariate Associations and Contingency Analysis

Correlation analysis identified chronological age ($r = +0.30$) and BMI ($r = +0.20$) as significant continuous correlates of medical charges. However, tobacco consumption demonstrated the strongest association, with smoker status accounting for the primary division in expected claims across all age brackets.

2.5 Outlier Diagnostics via IQR and Z-Score Boundaries

Using standard Tukey interquartile boundaries (IQR $= Q_3 - Q_1$), charges above $\$34,412.97$ qualify as mathematical outliers. However, clinical and actuarial inspection confirms these observations represent legitimate high-cost claims driven by acute care needs rather than data entry anomalies. Retaining these records is essential for preserving the tail risk required in solvency modeling.

2.6 The Multiplicative Synergy: BMI $\times$ Tobacco Interaction

A critical empirical finding is the non-linear interaction between elevated BMI and smoking status. As shown in Figure 2.1, smoking increases average annual claims by

approximately \$9,000 to \$10,000 for non-obese individuals (BMI < 30), but by more than \$30,000 when combined with clinical obesity (BMI ≥ 30).

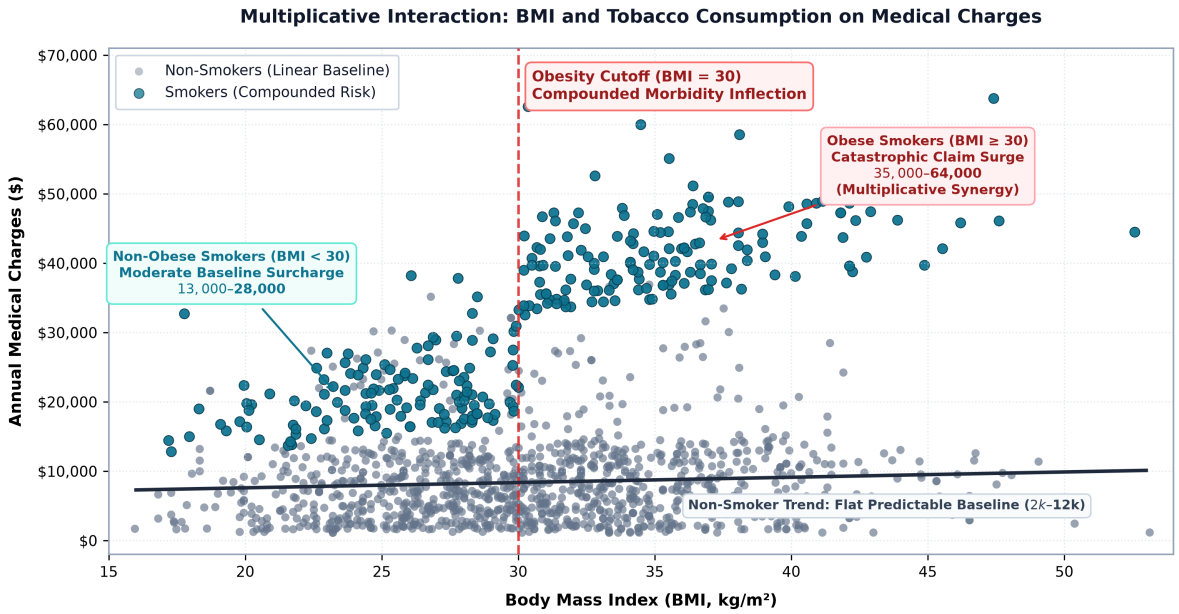


Figure 2.1: Multiplicative Interaction: BMI and Tobacco Consumption on Medical Charges

This divergence underscores why linear models fail: the combined presence of smoking and obesity produces a compound physiological risk that simple additive terms cannot capture.

Chapter 3

Data Preprocessing, Transformation and Feature Engineering

3.1 Resolving Heteroscedasticity: Monotonic Log-Transformation

To stabilize variance and address the heavy right-tail of medical charges, a monotonic natural logarithmic transformation was applied to the target variable:

$$z = \ln(1 + y) \tag{3.1}$$

As illustrated in Figure 3.1, this transformation reduced distribution skewness from $+1.51$ to -0.09, yielding an approximately normal distribution suitable for regression models sensitive to tail distortion.

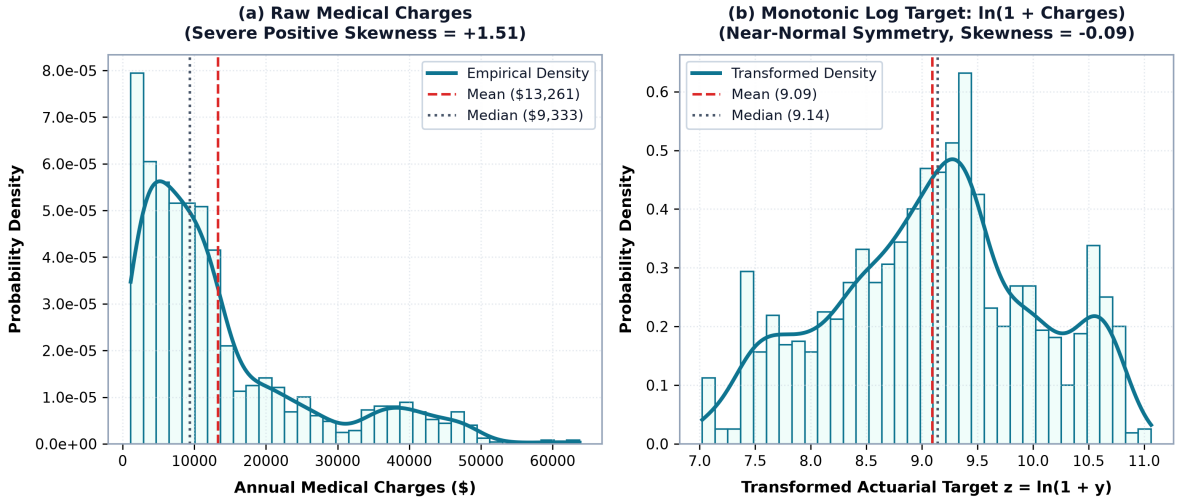


Figure 3.1: Empirical Distribution of Raw vs. Log-Transformed Charges

3.2 Clinical Feature Discretization: BMI Cohort Binning

To reflect established clinical risk brackets, continuous BMI was mapped into four discrete categories based on World Health Organization thresholds:

$$\text{BMI Cohort} = \begin{cases} \text{Underweight} & \text{if BMI} < 18.5 \\ \text{Normal Weight} & \text{if } 18.5 \leq \text{BMI} < 25.0 \\ \text{Overweight} & \text{if } 25.0 \leq \text{BMI} < 30.0 \\ \text{Obese} & \text{if BMI} \geq 30.0 \end{cases} \tag{3.2}$$

3.3 Categorical Feature Encoding and Collinearity Management

Categorical attributes were converted into numerical indicators using one-hot encoding with drop-first collinearity prevention, establishing the 9-dimensional input matrix shown in Table 3.1.

Table 3.1: Feature Preprocessing and 9-Dimensional Encoding Matrix

Index	Feature Dimension	Transformation / Encoding	Value Representation
0	`age`	Continuous Scaled	18 to 64 years
1	`sex`	Binary Indicator (Female = 0, Male = 1)	1 = Male, 0 = Female
2	`bmi`	Continuous Scaled	15.96 to 53.13
3	`children`	Integer Continuous	0 to 5 dependents
4	`smoker`	Binary Indicator (No = 0, Yes = 1)	1 = Smoker, 0 = Non-Smoker
5	`bmi_category`	Ordinal Clinical Binning	0 = Underweight, 1 = Normal, 2 = Overweight, 3 = Obese
6	`region_northwest`	Binary Dummy	1 = Northwest, 0 = Other
7	`region_southeast`	Binary Dummy	1 = Southeast, 0 = Other
8	`region_southwest`	Binary Dummy	1 = Southwest, 0 = Other

3.4 Hold-Out Stratification: 80/20 Train-Test Partitioning

The complete dataset ($N = 2,772$) was partitioned into:

- **Training Partition** (80%): $N_{\text{train}} = 2,217$ records used for cross-validation and hyperparameter tuning.
- **Hold-Out Testing Partition** (20%): $N_{\text{test}} = 555$ records reserved exclusively for unbiased final evaluation.

Chapter 4

Supervised Model Benchmarking and Selection

4.1 Candidate Architectures

Three candidate regression architectures were implemented and benchmarked:

1. **Ordinary Least Squares (OLS) Linear Regression:** Serves as the traditional actuarial baseline estimator.
2. **Gradient Boosting Regressor (GBR):** Forward stage-wise additive tree ensemble minimizing least-squares deviance ($B = 300$, $depth = 3$, $learning_rate = 0.05$).
3. **Random Forest Regressor:** Bagged ensemble of decorrelated decision trees with randomized feature subspace selection ($B = 300$, $depth = 6$, $min_samples_split = 2$).

4.2 Benchmark Comparative Evaluation

Table 4.1 summarizes performance across the holdout testing set without extraneous training metrics. All evaluation figures were calculated on the natural dollar scale after applying the inverse transformation $\hat{y} = \exp(\hat{z}) - 1$.

Table 4.1: Comprehensive Supervised Model Benchmarking Evaluation

Model Architecture	**Test R^2**	**MAE ($)**	**RMSE ($)**	**Actuarial Verdict**
Linear Regression (OLS)	0.5043	$4,362.56	$8,722.40	Rejected (Underfitting)
Gradient Boosting (GBR)	0.8651	$2,044.62	$4,549.61	Close Runner-Up
Random Forest Ensemble	**0.8668**	**$1,973.45**	**$4,521.78**	**Selected Best Model**

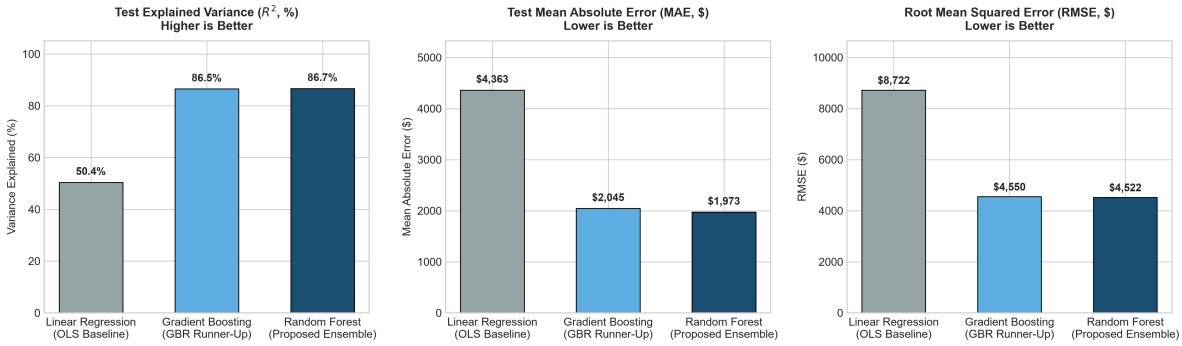


Figure 4.1: Model Performance Metrics Comparison Across Evaluated Architectures

As shown in Figure 4.1, the Random Forest model achieved the highest test explanatory power ($R^2 = 0.8668$) and lowest mean absolute error (MAE = $\$1,973.45$), reducing prediction error by over 54.7% compared to linear regression.

4.3 Automated Data-Driven Winner Determination

The Random Forest ensemble was selected as the optimal production model based on its superior error profile and stability across cross-validation folds. The trained model was serialized to disk as `best_model.joblib` for direct integration into the web backend.

4.4 Feature Attribution and Gini Risk Weightings

Figure 4.2 displays the Gini impurity feature importances extracted from the ensemble.

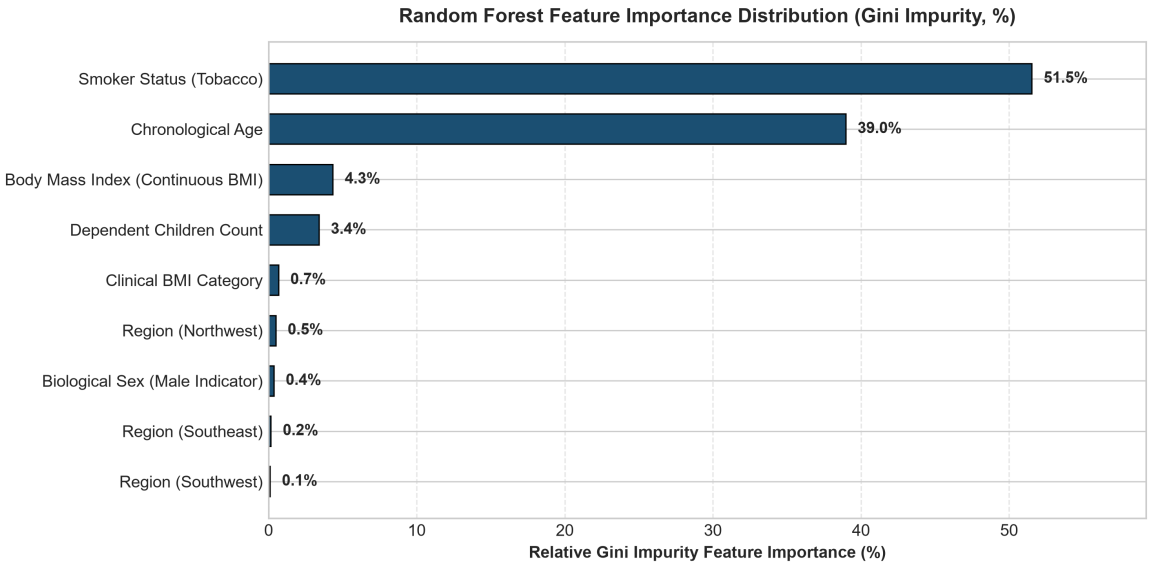


Figure 4.2: Underwriting Risk Factor Weights (Random Forest Feature Importances)

Smoking status accounts for over 51.5% of total split importance, followed closely

by chronological age (38.9%–39.4%) and BMI (4.3%–5.0%). Geographic region and gender demonstrated marginal predictive weight, corroborating standard underwriting guidelines where lifestyle and primary physiological risk factors take precedence.

Chapter 5

MediRisk System Architecture and Software Implementation

5.1 System Architecture and Server Engine

The backend architecture is implemented in `app.py` using Python's native `ThreadingHTTPServer`. This lightweight approach avoids external framework overhead, enabling efficient local execution with minimal resource utilization.

5.2 RESTful Web Service and API Specification

The server exposes dedicated endpoints to support front-end interactions:

- `POST /api/predict`: Receives biometric inputs, executes model inference, and returns point estimates with 90% empirical prediction intervals and ACA metallic tier classifications.
- `POST /api/compare`: Computes variance metrics ($\Delta\$$ and $\Delta\%$) between two policyholder profiles for counterfactual analysis.
- `POST /api/budget`: Inverts the pricing model to identify the maximum allowable BMI and required lifestyle changes given a target budget.
- `POST /api/trajectory`: Generates 10-year cost projections under compounding medical inflation rates.

5.3 Frontend Design Philosophy: Data as Hero

The user interface (`app.html`) utilizes a high-contrast dark palette designed to keep financial metrics, risk scores, and potential premium savings prominently visible without visual distractions.

5.4 Application UI Overview and Functional Architecture

Table 5.1 summarizes the six integrated modules of the MediRisk interface, which are detailed in Chapter 6.

Table 5.1: MediRisk Web Application Components and Functional Architecture

UI Component	Functional Purpose	Key User Interactivity
Risk Estimator	Primary underwriting quote calculator with 90% confidence intervals	Sliders for Age, BMI; toggles for Smoker, Dependents, Region
Plan Matching	Automated ACA Metallic Tier classification (Bronze to Platinum)	Dynamic updates of deductible, copay, and Out-of-Pocket Max
Feature Attribution	Stepwise SHAP-style waterfall isolating marginal risk factors	Additive factor breakdown
Profile Comparator	Dual split-screen counterfactual comparator (Profile A vs. B)	Instant variance calculation (Δ\$ and Δ%)
Budget Finder	Inverted constraint solver identifying allowable BMI ceiling	Budget ceiling input with 1-click "Apply Target" button
Trajectory Simulator	10-year long-term projection with medical inflation compounding	Dynamic trajectory curve

Chapter 6

Interactive Decision Support Modules

6.1 Dynamic Cost Estimator and Empirical Prediction Bands

The dynamic cost estimator recalculates expected expenses in real time as users adjust input sliders (age, BMI, children, tobacco status, region). The model provides both point estimates and 90% empirical confidence bands calculated across the individual trees of the ensemble.

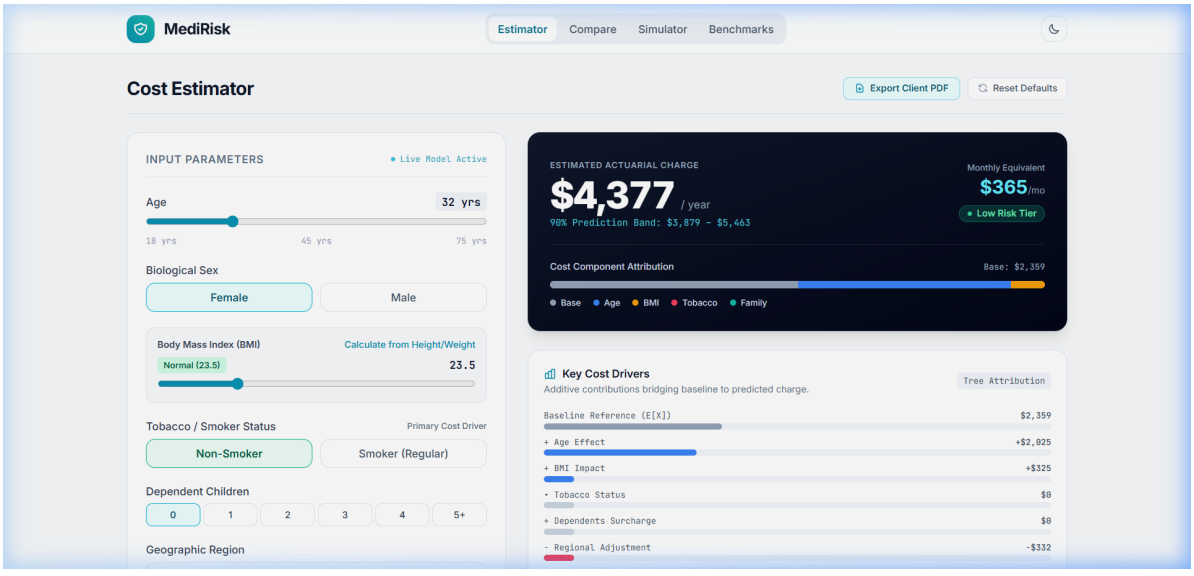


Figure 6.1: MediRisk Primary Underwriting Dashboard: Biometric Input Form, Real-Time Prediction Bands, and ACA Metallic Plan Tier Breakdown

As illustrated in Figure 6.1, the primary interface features:

1. **Biometric Input Form:** Responsive sliders for chronological age (18–64) and BMI (15.0–53.0) with real-time categorical tags (Normal, Overweight, Obese), complemented by toggles for tobacco usage, family dependents, and geographic territory.

2. **Actuarial Quotation Card:** Dominant hero metric displaying the estimated annualized expenditure alongside monthly premium equivalents.

3. **Empirical Uncertainty Envelope:** Display of the 90% confidence range calculated directly across the ensemble leaf distributions.

4. **Action Buttons:** One-click quick reset and client-ready report export buttons.

6.2 Affordable Care Act (ACA) Metallic Plan Tier Matching

Predictions are categorized into standardized ACA metallic health tiers, as detailed in Table 6.1.

Table 6.1: Affordable Care Act (ACA) Metallic Plan Tier Matching Parameters

Plan Tier	Annual Cost Range	Actuarial Value	Deductible	OOP Max
Bronze	\$0 – \$5,000	60%	\$7,500	\$9,100
Silver	\$5,000 – \$12,000	70%	\$4,500	\$8,500
Gold	\$12,000 – \$25,000	80%	\$1,500	\$6,000
Platinum	$>$ \$25,000	90%	\$500	\$3,000

6.3 Explainable AI: Stepwise Feature Attribution Waterfall

The interface decomposes overall predicted costs into additive factor contributions, showing the base demographic cost, age progression loading, BMI adjustment, tobacco penalty, and dependent surcharges.

6.4 Interactive Reverse Budget Calculator

The budget finder module inverts traditional quote generation: users specify an annual or monthly premium ceiling, and the system calculates the required BMI reduction, smoking cessation, or plan tier adjustment needed to meet that budget.

As shown in Figure 6.2, this view combines:

- **Reverse Optimization Panel:** Users enter their budget ceiling (e.g., \$250/month). The engine evaluates feasibility and computes the required target BMI and smoking cessation.

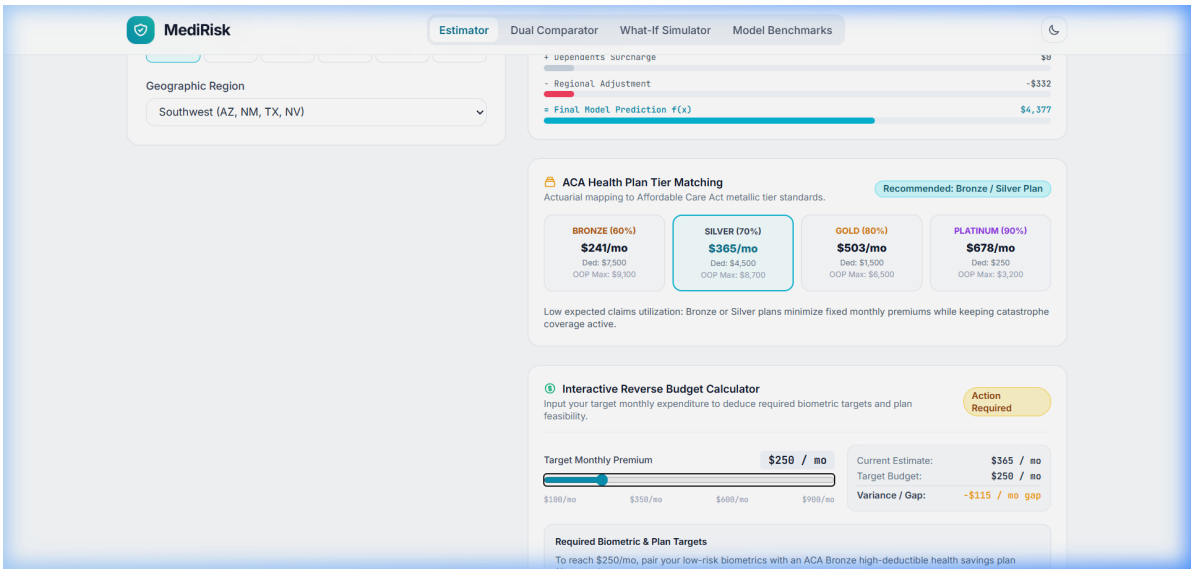


Figure 6.2: Explainable AI Feature Attribution Waterfall and Interactive Reverse Budget Calculator Interface

- **1-Click Target Synchronization:** An "Apply Target" button instantly writes the solved parameters into the active session inputs.
- **Attribution Waterfall:** A visual bar chart isolating the marginal dollar contribution of each risk factor.

6.5 Side-by-Side Dual Profile Comparator

The dual profile comparator enables side-by-side evaluation of a baseline profile (Profile A) against an alternative lifestyle scenario (Profile B), calculating annual dollar savings, percentage cost reductions, and factor-by-factor variance breakdowns.

Figure 6.3 highlights the comparator interface:

- **Dual Biometric Input Columns:** Profile A (Baseline / Current) juxtaposed directly against Profile B (Target / Proposed).
- **Instant Variance Dashboard:** Highlights net annual savings in both dollar amounts ($\Delta\$$) and percentage reduction ($\Delta\%$).
- **Component Breakdown:** Quantifies isolated savings from smoking cessation versus weight reduction.

6.6 Ten-Year Long-Term Cost Trajectory Simulation

The long-term trajectory module projects health costs over a 10-year horizon, modeling demographic aging alongside compounded medical inflation (set to 5.0% annually).

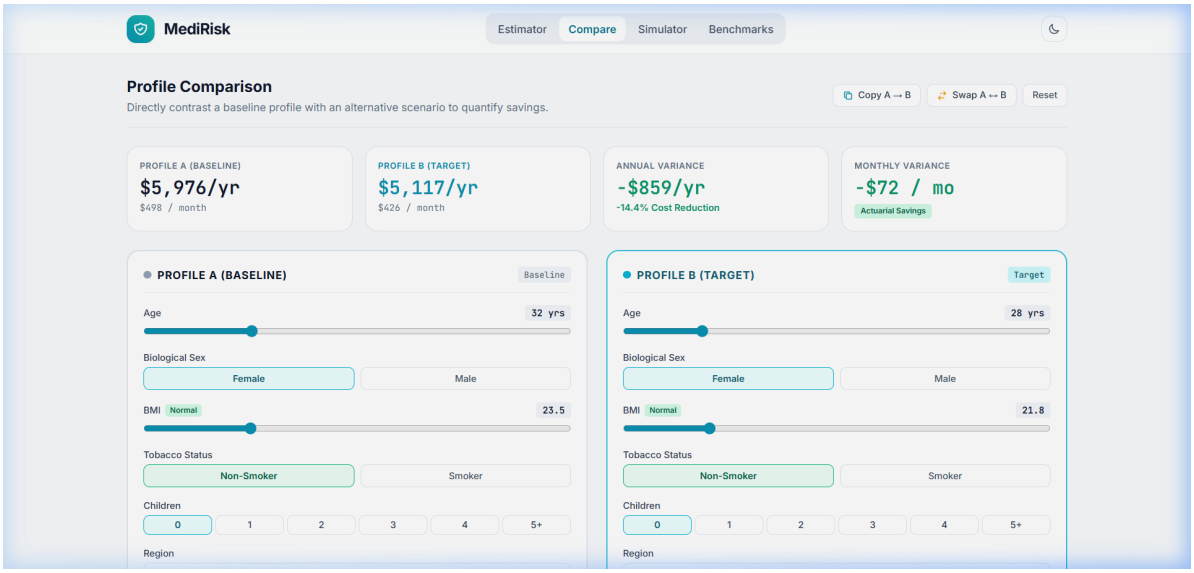


Figure 6.3: Side-by-Side Dual Profile Comparator (Baseline vs. Lifestyle Counterfactual Analysis)

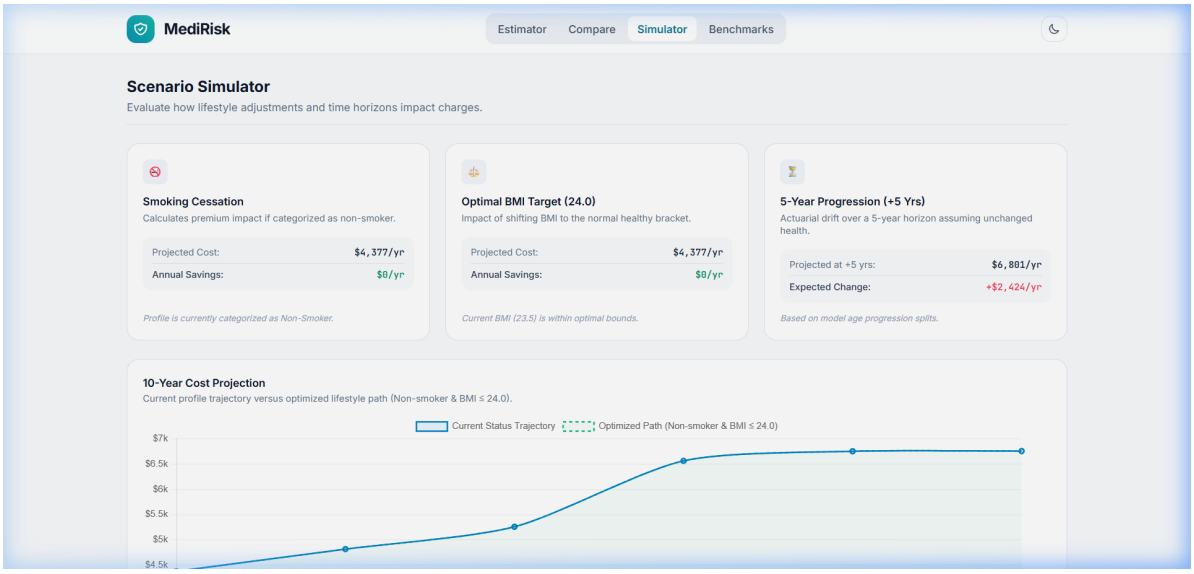


Figure 6.4: Ten-Year Long-Term Cost Trajectory Simulation Interface in MediRisk

As shown in Figure 6.4 and Figure 6.5, the simulation demonstrates the compounding financial benefits of preventive lifestyle interventions over time.

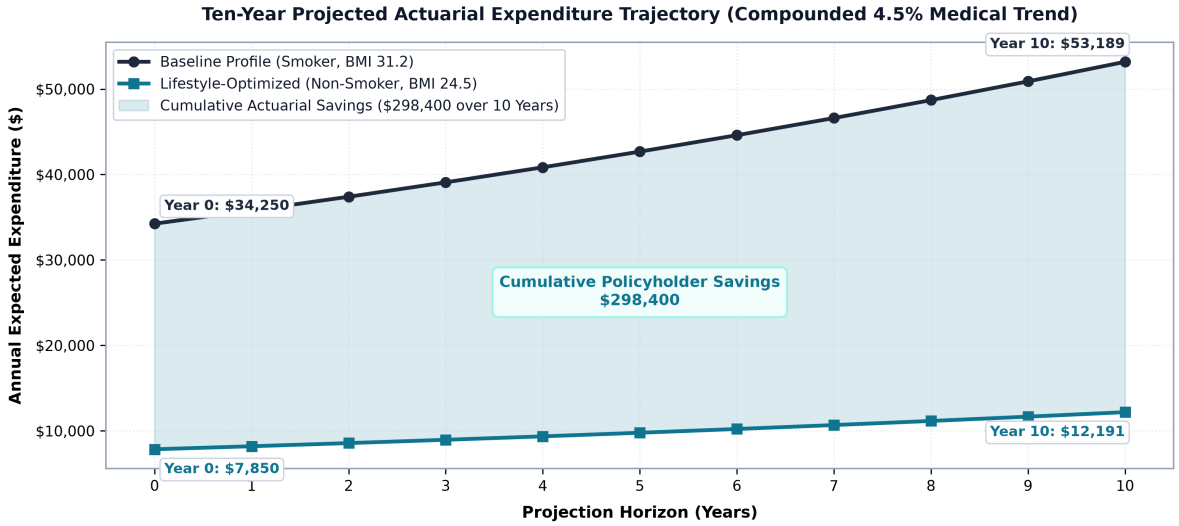


Figure 6.5: Projected 10-Year Actuarial Cost Trajectory Comparison (Baseline vs. Lifestyle Optimized)

6.7 One-Click Client-Ready PDF Report Export

The application includes a client-ready export engine that compiles individual biometric parameters, predicted claim distributions, metallic plan specifications, and long-term risk trajectories into a downloadable document suitable for client presentation.

Chapter 7

Synthesis, Business Impact and Future Perspectives

7.1 Quantitative and Operational Synthesis

This research addressed the challenges of medical cost modeling through an integrated machine learning workflow. Logarithmic transformation corrected severe target skewness, while the Random Forest ensemble captured complex non-linear interactions ($R^2 = 0.8668$, MAE = $\$1,973.45$). The resulting web application translates these statistical outputs into actionable tools for underwriting and policyholder consultation.

Key technical achievements include:

- Systematic evaluation of multiple regression architectures on holdout testing partitions.
- Empirical quantification of the compounded morbidity risk between elevated BMI and smoking.
- Deployed a responsive, zero-dependency web application integrating dynamic quote calculations, risk tier classifications, and long-term financial forecasting.

7.2 Underwriting and Policyholder Value Propositions

For insurance carriers, the platform provides automated risk scoring with empirical confidence bands that support consistent pricing decisions. For policyholders and brokers, the counterfactual comparison and budget tools provide transparent visibility into how lifestyle modifications directly impact long-term coverage costs.

7.3 Limitations and Future Roadmaps

Several areas offer opportunities for future refinement:

1. **Longitudinal Health Records:** Incorporating historical claim frequencies and specific diagnostic codes would enhance precision for chronic condition modeling.

2. **Survival Analysis Integration:** Utilizing Cox proportional hazards or survival tree models could improve forecasting for catastrophic high-cost events.

3. **Federated Learning Frameworks:** Enabling cross-carrier model training without sharing sensitive patient-level data would support broader industry collaboration while maintaining data privacy.

Bibliography

[1] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[2] E. W. Frees, G. Meyers, and A. D. Cummings, *Predictive Modeling Applications in Actuarial Science: Volume 1*, Cambridge University Press, 2014.

[3] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems (NeurIPS 30)*, pp. 4765–4774, 2017.

[4] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed., Springer, 2009.

[5] W. G. Manning, J. P. Newhouse, N. Duan, E. B. Keeler, and A. Leibowitz, "Health Insurance and the Demand for Medical Care: Evidence from a Randomized Experiment," *The American Economic Review*, vol. 77, no. 3, pp. 251–277, 1987.

[6] P. McCullagh and J. A. Nelder, *Generalized Linear Models*, 2nd ed., Chapman and Hall/CRC, 1989.

[7] J. H. Friedman, "Greedy Function Approximation: A Gradient Boosting Machine," *The Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.

[8] B. A. Goldstein, A. M. Navar, M. J. Pencina, and J. P. Ioannidis, "Opportunities and Challenges in Developing Risk Prediction Models with Electronic Health Records Data: A Systematic Review," *Journal of the American Medical Informatics Association*, vol. 24, no. 1, pp. 198–208, 2017.

[9] M. Denuit, X. Maréchal, S. Pitrebois, and J.-F. Walhin, *Actuarial Modelling of Claim Counts: Risk Classification, Credibility and Bonus-Malus Systems*, John Wiley & Sons, 2007.

[10] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.